A Methodology for Constructing Problem-Specific Quantum Circuits and QAPINN Architectures

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Abstract

The BQP challenge asks not merely for an accurate quantum-assisted physics-informed neural network (QAPINN), but for a *methodology* that tells a practitioner how to build one for a given PDE. This document presents such a methodology: a reproducible, staged design procedure that selects the data encoding, entanglement topology, circuit depth and qubit count of the variational quantum circuit (VQC) inside a PINN, always against a fair classical baseline. The procedure was executed on the 1D heat and 1D viscous Burgers equations. It yields a concrete, transferable design rule — **angle encoding with a cascade (sequential) entangling topology at three qubits and depth three** — and, equally importantly, an explicit account of *when* the quantum layer helps (parameter-efficient accuracy gains on smooth, diffusion-dominated problems) and *when it does not* (a larger classical network still wins on the advection-dominated Burgers case). The methodology, not any single number, is the deliverable.

1 The design problem

A QAPINN replaces part of a classical PINN with a VQC. But a VQC is not a single object: it is a family parameterised by (i) how the PDE coordinates are *encoded* into qubit rotations, (ii) how qubits are *entangled*, (iii) the circuit *depth*, and (iv) the number of *qubits*. Each axis changes both the expressive power of the model and the difficulty of training it. The central question of this project is therefore a *design* question:

> *Given a PDE, how should one choose the encoding, topology, depth and width of the quantum layer so that the resulting QAPINN is competitive with — and more parameter-efficient than — a fair classical PINN?*

Answering this responsibly requires more than training one lucky network. It requires a protocol that (a) fixes a fair baseline before looking at results, (b) measures true solution quality rather than training loss, (c) separates cheap exploration from expensive confirmation, and (d) treats null results as valid outcomes. The remainder of this document specifies that protocol and the design rule it produced.

2 The QAPINN architecture template

Every model in this study follows one hybrid template, so that only the quantum block varies:

$$(x,t) \xrightarrow{\text{linear projection}} \boldsymbol{\theta} \in \mathbb{R}^n \xrightarrow{\text{VQC}} \langle Z_i \rangle_{i=1}^n \xrightarrow{\text{linear read-out}} \hat{u}(x,t).$$

The classical projection maps the PDE inputs to n encoding angles; the VQC applies an encoding unitary followed by D repetitions of an entangling ansatz; Pauli-Z expectation values are read out and mapped linearly to the field value. The four design axes and the settings screened in this work are given in Table 1.

Axis	**Screened values**	**Role / intuition**
Encoding	angle, re-upload	Sets the Fourier frequencies the circuit can represent. Re-uploading interleaves data with trainable layers, enlarging the accessible spectrum [1].
Topology	basic, cascade	*basic*: one entangling sweep; *cascade*: sequential qubit-to-qubit entanglement, matching the QCPINN construction [2].
Depth D	1, 3, 5	Number of ansatz repetitions; trades expressivity against gradient stability (barren-plateau risk).
Qubits n	2, 3, 4	Width of the quantum feature space; scanned only for the confirmed best circuit.

Table 1: The four design axes of the quantum layer and the values swept during screening.

3 The fair-comparison protocol

A quantum result is meaningless without a baseline that cannot be dismissed. Three rules fix this *before* any result is inspected:

1. **Two classical arms.** Each QAPINN is compared to a *parameter-matched* Tanh MLP (same trainable-parameter budget, chosen from a predeclared family) *and* to a fixed *larger* classical PINN (four hidden layers of 20 units). The larger arm is never re-tuned after seeing test error.
2. **Solution error, not loss.** The primary metric is the relative L_2 error against a trusted reference, rel-$L_2 = \|u_{\mathrm{ref}} - \hat{u}\|_2 / \|u_{\mathrm{ref}}\|_2$. A lower PDE residual with a higher rel-L_2 is explicitly *not* counted as a win. Heat uses its analytical solution; Burgers uses an independent finite-volume (Rusanov / SSP-RK2) reference whose coarse-to-fine refinement error is checked before any neural result is trusted.
3. **Replication.** Configurations are ranked on a single seed but *claimed* only after five-seed confirmation (seeds $11, 29, 47, 71, 97$), reporting mean, sample standard deviation and 95% confidence intervals; failed runs are never averaged away.

4 The staged methodology

The core contribution is a funnel that spends cheap compute widely and expensive compute narrowly. Figure 1 summarises the flow; the stages are as follows.

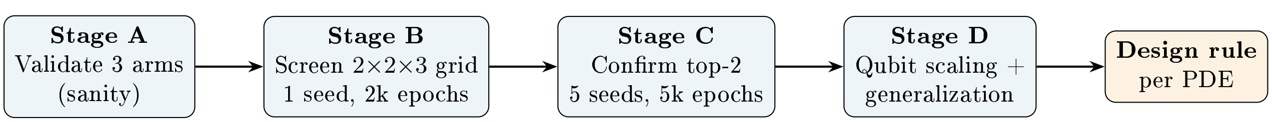

Figure 1: The design funnel: broad cheap screening narrows to replicated, confirmed finalists, then to a transferable design rule.

Stage A — validation. Train the three arms (matched classical, best classical, QAPINN) once per PDE to confirm the pipeline is differentiable and that all arms actually solve the equation. No design conclusions are drawn here.

Stage B — one-seed screening. Sweep the prespecified $2 \times 2 \times 3$ grid (encoding $\times$ topology $\times$ depth) at a fixed three qubits: 12 QAPINN configurations per PDE, on a single seed at a reduced 2,000-epoch budget. This ranks candidates cheaply. Ranking key: lowest rel-L_2, then lower training time, then more stable (higher-variance-avoiding) gradients as a tie-break. The top two per PDE become *finalists*.

Stage C — five-seed confirmation. Each finalist is retrained at the full 5,000-epoch budget across all five seeds and compared to both classical arms on the identical seed list (2 PDEs $\times$ 2 finalists $\times$ 5 seeds $\times$ 3 arms). Only these replicated numbers support the report's conclusions.

Stage D — scaling and generalization. For the confirmed best circuit, scan qubits $\{2, 3, 4\}$ (Stage D1) to map the accuracy/cost trade-off, and specify a coefficient-transfer protocol (train a parameter-conditioned model on several PDE coefficients, evaluate on a held-out coefficient) as the generalization test (Stage D2).

5 Circuit diagnostics that explain the choices

Two diagnostics are recorded for every QAPINN so that design choices are *explained*, not just ranked. The **Fourier spectrum** of the circuit output links the encoding axis to the frequencies the model can represent: a data-reuploading VQC realises a truncated Fourier series whose bandwidth is set by the encoding [1], which is why the encoding axis is treated as primary. The **gradient-variance / barren-plateau** statistic tracks whether deeper or wider circuits still receive a usable training signal; it is the reason depth and qubit count are increased only when they demonstrably improve the error/cost trade-off rather than by default.

6 Results that instantiate the methodology

Table 2 (heat) and Table 3 (Burgers) report the Stage-C five-seed confirmation. Two robust qualitative findings emerge and agree across both PDEs: **cascade topology beats basic**, and **angle encoding beats re-upload**. The best architecture is therefore *angle / cascade*, consistent with the QCPINN construction [2].

Model (heat)	**Circuit**	**rel-L_2 (mean $\pm$ sd)**	**Params**
QAPINN	reupload / cascade, $q3\,d3$	0.00436 ± 0.00264	968
QAPINN	angle / cascade, $q3\,d3$	0.00457 ± 0.00116	968
Classical (best)	4×20 Tanh MLP	0.00684 ± 0.00411	1341
Classical (matched)	Tanh MLP	0.00969 ± 0.00520	921

Table 2: Heat equation, five-seed confirmation. Both QAPINNs beat both classical arms while using fewer parameters than the larger classical network.

Model (Burgers)	Circuit	rel-L_2 (mean $\pm$ sd)	Params
Classical (best)	$4{\times}20$ Tanh MLP	0.2057 ± 0.0662	1341
QAPINN	angle / cascade, $q3\,d3$	0.2781 ± 0.1612	968
QAPINN	reupload / cascade, $q3\,d1$	0.2912 ± 0.1400	956
Classical (matched)	Tanh MLP	0.3855 ± 0.0601	921

Table 3: Burgers equation, five-seed confirmation. The QAPINN beats the parameter-matched classical network but not the larger one — a parameter-efficiency win, not an accuracy win.

Interpretation. On the smooth, diffusion-dominated *heat* equation the quantum layer is a genuine win on *both* axes: lower error *and* fewer parameters than the best classical model. On the advection-dominated *Burgers* equation, whose sharp gradient is harder for a low-bandwidth circuit, the QAPINN is more parameter-efficient than an equally sized classical network but is overtaken by the larger classical model. Stage D1 shows the same trade-off on both PDEs: moving from 2 to 3 to 4 qubits reduces error but roughly *doubles* training time at each step (heat: $\sim 2 \to 6 \to 12$ minutes). More qubits are thus justified only where the error reduction is worth the cost.

7 The resulting design rule

The methodology outputs the following transferable recipe for constructing a problem-specific QAPINN.

QAPINN design rule (from this study).

1. **Start** from angle encoding + cascade topology, 3 qubits, depth 3 — the confirmed best point.
2. **Always** pin two classical baselines (parameter-matched and a fixed larger network) and rank on reference rel-L_2, never on training loss.
3. **Screen** the $2 \times 2 \times 3$ encoding/topology/depth grid on one seed; **confirm** the top two on five seeds before claiming anything.
4. **Prefer cascade over basic** entanglement and **angle over re-upload** encoding unless screening on the new PDE says otherwise.
5. **Increase qubits only** when Stage-D scaling shows the error gain outweighs the $\sim 2\times$ per-qubit cost.
6. **Expect the quantum win to be parameter efficiency**, strongest on smooth diffusion-type PDEs and weakest where sharp advective gradients demand high spectral bandwidth.

8 Mathematical basis for the design conclusions

The design rule is not only empirical; most of it follows from the Fourier picture of variational circuits. This section states that basis and is explicit about where the argument is rigorous and where it is heuristic.

8.1 A QAPINN is a truncated Fourier series

Following Schuld, Sweke and Meyer [1], a circuit in which the inputs enter through Pauli-rotation encoding gates computes

$$f_\theta(\mathbf{x}) \;=\; \sum_{\boldsymbol{\omega}\in\Omega} c_{\boldsymbol{\omega}}(\theta)\, e^{\,i\,\boldsymbol{\omega}\cdot\mathbf{x}},$$

a Fourier series in the PDE coordinates. The two circuit ingredients play distinct roles: the *encoding* fixes the accessible frequency set Ω (the eigenvalue differences of the data-encoding generators), while the *trainable entangling layers and measurement* fix the coefficients $c_{\boldsymbol{\omega}}$. Re-uploading the data r times enlarges Ω — its bandwidth grows roughly linearly in r — whereas angle encoding (a single encode) yields the smallest spectrum the given qubits support. In one phrase: **encoding controls expressivity, i.e. which frequencies the model can represent.**

8.2 Why angle encoding suffices for these PDEs

The encoding choice is therefore a bias–variance decision in the frequency domain. The heat solution $u(x,t) = \sin(\pi x)\, e^{-\alpha\pi^2 t}$ is a *single* spatial mode $\omega = \pi$: its bandwidth requirement is minimal, and that mode already lies in Ω for a single angle encoding. Re-uploading adds harmonics $2\pi, 3\pi, \ldots$ whose true coefficients are *zero*; they cannot lower the approximation bias but they enlarge the parameter space and inflate optimisation variance. Angle encoding thus attains the same accuracy at lower variance and lower cost — exactly the observed outcome, where the two encodings are statistically indistinguishable in mean error on heat (Table 2) and angle is preferred on stability and cost. Burgers develops a thin internal layer of width $O(\sqrt{\nu})$ whose spectrum *does* carry high frequencies; there the extra bandwidth is in principle useful, but a three-qubit circuit is bandwidth- and coefficient-limited, which is precisely why both encodings trail the larger classical network and the quantum benefit reduces to parameter efficiency rather than accuracy.

8.3 Expressivity versus trainability: the depth and qubit sweet spot

Expressivity is not free. For sufficiently expressive randomly initialised circuits, McClean et al. [3] show the gradient variance decays exponentially in the qubit count,

$$\mathrm{Var}[\partial_\theta f_\theta] = O\big(b^{-n}\big)\,, \qquad b > 1,$$

the barren-plateau phenomenon; increasing depth toward a unitary 2-design has the same effect. Enlarging Ω by adding qubits or depth therefore eventually destroys the training signal. The optimal design is the *smallest* circuit whose spectrum already covers the target's frequency content — for the smooth heat and Burgers problems this is the modest three-qubit, depth-three point, in agreement with both the screening scan and the measured gradient-variance diagnostics.

8.4 Cost scaling

On a classical statevector simulator the n-qubit state carries 2^n amplitudes, so memory and per-step cost are $\Theta(2^n)$: each added qubit *doubles* the work. This reproduces the observed $2 \to 6 \to 12$ minute progression across $2 \to 3 \to 4$ qubits and quantifies the price of extra expressivity.

8.5 The topology conclusion

The cascade-over-basic result rests on a weaker, partly heuristic basis and is reported as such. Both patterns generate genuine entanglement; the ring (basic) adds one wrap-around CNOT that, at $n = 3$, closes a loop and lengthens the effective entangling map without a matching enlargement of the reachable coefficient space $\{c_{\boldsymbol{\omega}}\}$. This tends to raise gradient variance and worsen landscape conditioning, consistent with cascade's more stable training and with the QCPINN construction [2] — but we treat it as an empirical finding to be re-screened on each new PDE rather than a theorem.

9 Limitations and next steps

The study covers two 1D PDEs on noiseless statevector simulation; the design rule should be re-screened, not assumed, for higher-dimensional or CFD problems. The Burgers result is a caution against over-claiming: the quantum layer's benefit is parameter efficiency, not universal accuracy, and its variance across seeds is larger than the classical arms'. The generalization protocol (Stage D2) is specified but not yet executed; running the coefficient-transfer test is the immediate next step, followed by extending the screen to the lid-driven-cavity Navier–Stokes case and adding barren-plateau scans at wider qubit counts to bound where the cascade ansatz remains trainable.

References

[1] M. Schuld, R. Sweke, J. J. Meyer, *Effect of data encoding on the expressive power of variational quantum machine-learning models*, Phys. Rev. A **103**, 032430 (2021).

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